Math Club Proof Workshop: Direct Proofs & Divisibility

Worksheet & Lab Problem Set

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1. Reference Definitions & Rules

Definition 1 (Even)**.** An integer n is **even** if there exists an integer $k \in \mathbb{Z}$ such that $n = 2k$.

Definition 2 (Odd)**.** An integer n is **odd** if there exists an integer $k \in \mathbb{Z}$ such that $n = 2k + 1$.

Definition 3 (Divisibility)**.** For integers a and b with $a \neq 0$, we say a **divides** b (written $a \mid b$) if there exists an integer $k \in \mathbb{Z}$ such that $b = ak$.

Integer Arithmetic Rule. The sum, difference, and product of any two integers is always an integer (closure under $+, -, \times$).

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2. Lecture Demo

Theorem 1. *The sum of an even integer and an odd integer is an odd integer.*

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3. Proof Lab Problems

Note: Use separate scratch paper for non-linear discovery. Once an officer checks your logic, write your final proof in complete English prose below.

Theorem 2 (Warmup)**.** *If n is an even integer, then $n^2 + 3n$ is an even integer.*

Formal Proof (Complete English Sentences)

Theorem 3 (Medium)**.** *For all integers a, b, and c, if $a \mid b$ and $a \mid c$, then $a \mid (3b - 2c)$.*

Formal Proof (Complete English Sentences)

Theorem 4 (Challenge)**.** *If n is an odd integer, then $n^2 - 1$ is divisible by 8.*

Formal Proof (Complete English Sentences)